EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2025-26

Course: Multidisciplinary Minor
Course code: MDC401
Year: SE SEM: IV

Experiment No. 5
AIM: Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.
Name: Ashish Ramesh Patel
Roll Number: 55
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty : Dr. Yogesh Pingle

Aim: Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.

Code:

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Experiment 5: Weather App</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

   <div class="container">
      <h2>Real-Time Weather</h2>

      <div class="input-group">
         <input type="text" id="cityInput" placeholder="Enter city name...">
         <button onclick="fetchWeather()">Search</button>
      </div>

      <div id="weatherDisplay" class="weather-info">
         <p>Enter a city to see the weather.</p>
      </div>
   </div>

   <script src="script.js"></script>
</body>
</html>
```

script.js:

```
function fetchWeather() {
    const city = document.getElementById("cityInput").value.trim();
    const displayArea = document.getElementById("weatherDisplay");

    if (city === "") {
        displayArea.innerHTML = "<p style='color:red;'>Please enter a city name.</p>";
        return;
    }

    // Your actual API key is placed here as a string
```

```
    const apiKey = "2fb5e266e1d779d2c5c9de39a0562d9c";

    // The API URL uses the dynamic 'city' variable and 'apiKey' variable
    const url =
`https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKey}&units=met
ric`;

    displayArea.innerHTML = "<p>Loading data...</p>";

    // Making the AJAX request using the Fetch API
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    fetch(url)
        .then(response => {
            if (response.status === 401) {
                throw new Error("API Key is invalid or still activating (takes 1-2 hours).");
            }
            if (response.status === 404) {
                throw new Error("City not found. Check your spelling.");
            }
            if (!response.ok) {
                throw new Error("Something went wrong with the network.");
            }
            return response.json(); // Convert the response to JSON
        })
        .then(data => {
            // Extracting the required data
            const temp = data.main.temp;
            const humidity = data.main.humidity;
            const conditions = data.weather[0].description;

            // Dynamically updating the HTML
            displayArea.innerHTML = `
                <p><strong>City:</strong> ${data.name}</p>
                <p><strong>Temperature:</strong> ${temp} °C</p>
                <p><strong>Humidity:</strong> ${humidity}%</p>
                <p><strong>Conditions:</strong> <span style="text-transform:
capitalize;">${conditions}</span></p>
            `;
        })
        .catch(error => {
            // Handle errors (like typing a fake city name)
            displayArea.innerHTML = `<p style='color:red;'>Error: ${error.message}</p>`;
        });
}
```
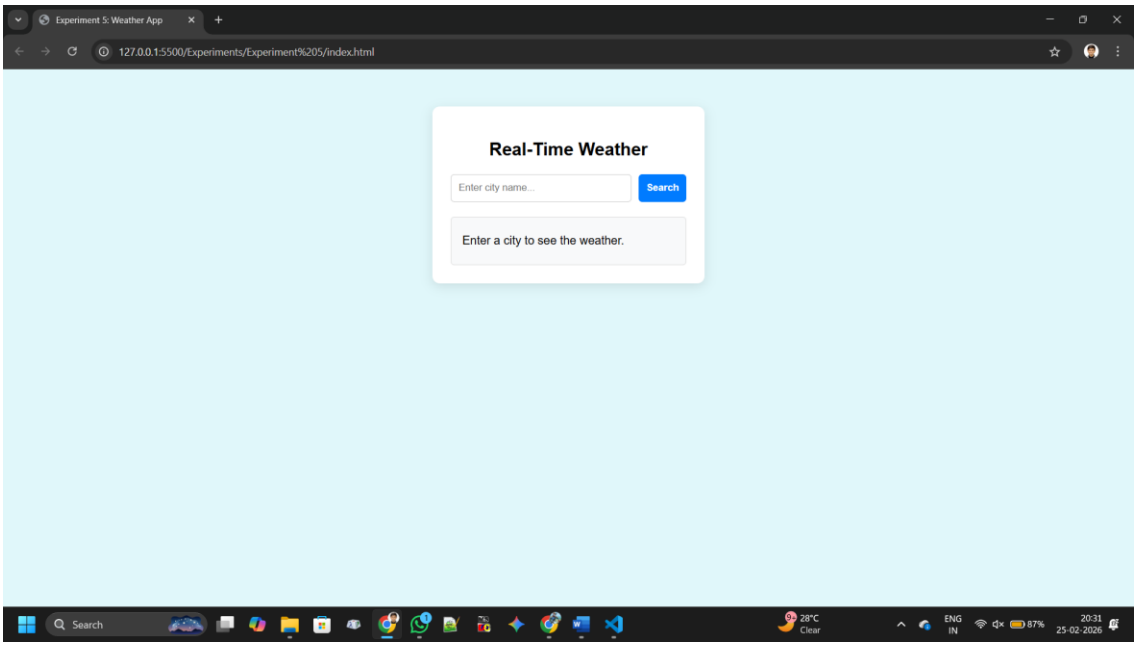
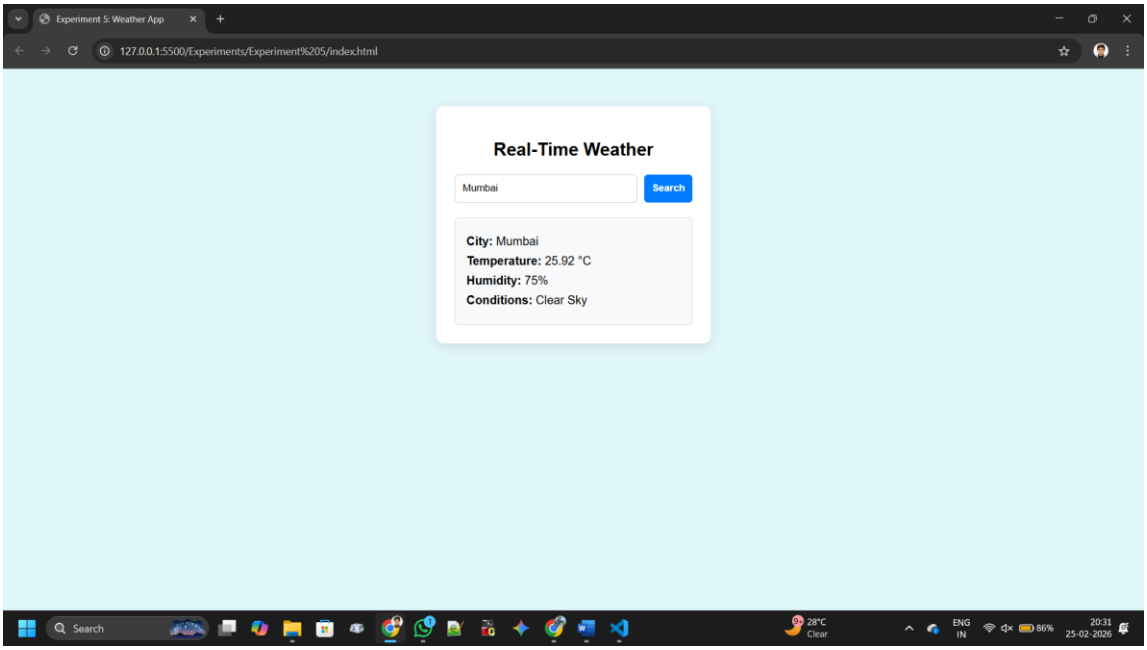

style.css:

```
body {
    font-family: Arial, sans-serif;
    background-color: #e0f7fa;
    display: flex;
    justify-content: center;
    padding-top: 50px;
    margin: 0;
}

.container {
    background: white;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0,0,0,0.1);
    width: 320px;
    text-align: center;
}

.input-group {
    display: flex;
    gap: 10px;
    margin-bottom: 20px;
}

input {
    padding: 10px;
    flex-grow: 1;
    border: 1px solid #ccc;
    border-radius: 5px;
}

button {
    padding: 10px;
    cursor: pointer;
    background-color: #007bff;
    color: white;
    border: none;
    border-radius: 5px;
    font-weight: bold;
}

button:hover {
    background-color: #0056b3;
}

.weather-info {
    background: #f8f9fa;
    padding: 15px;
    border-radius: 5px;
```

```
    border: 1px solid #ddd;
    text-align: left;
}

.weather-info p {
    margin: 8px 0;
    font-size: 16px;
}
```

Output: